

PH.D. QUALIFYING EXAM 2021: PROBABILITY

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You can take the exam at any location where you have the facility to scan or take pictures of your answers and convert them into pdf files. You are allowed to use any material of your choice prepared before the time of the exam, but use of online materials or consulting anyone else during the exam is not be permitted. If you have any questions during the exam, please email Professors Harry Crane (hcrane@stat.rutgers.edu) and Tirthankar Dasgupta (td370@rutgers.edu).

Please make sure to **handwrite** your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. **Typed answers are not permitted.**

Create two pdf files for your answers - one for each problem. Both the pdf files containing your answers should be emailed to td370@stat.rutgers.edu and hcrane@stat.rutgers.edu by 2:15 PM (or at an earlier time based on your location as agreed upon in your communication with the Graduate Directors). **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Good luck!

Problem 1

Note: This problem is about distributional properties of order statistics. Let $\mathbf{x} = (x_1, \dots, x_n)$ be a sample of real numbers. The *order statistics* for the sample $\mathbf{x}$ are the real numbers $x_{(1)}, \dots, x_{(n)}$ obtained by arranging the elements of $\mathbf{x}$ in increasing order, i.e.,

$$x_{(1)} < \dots < x_{(n)}.$$

For the first three parts of this problem, we let $U_1, \dots, U_n$ be i.i.d. Uniform $[0, 1]$ random variables and write

$$U_{(1)} < \dots < U_{(n)}$$

to denote their order statistics. Define

$$Y_1 := U_{(1)} - 0, \quad Y_2 := U_{(2)} - U_{(1)}, \quad \dots, \quad Y_i := U_{(i)} - U_{(i-1)}, \quad \dots, \quad Y_{n+1} := 1 - U_{(n)}$$

as the *spacings* of the order statistics.

Question 1: For $k = 2, 3, 4, \dots$, let

$$S_n^{(k)} := \sum_{i=1}^{n+1} Y_i^k$$

be the sum of the k th powers of the spacings for a sample of size n . Show that there exist constants $b_k > 0$ and $c_k > 0$ such that

$$(1) \quad n^{b_k} S_n^{(k)} \rightarrow_P c_k \quad \text{as } n \rightarrow \infty.$$

Identify the constants b_k and c_k , and point out that $b_2 < b_3 < \dots$.

Hint for Question 1: For each n , the joint distribution of spacings $Y_1, \dots, Y_{n+1}$ is the same as that of the random variables $X_1/T_{n+1}, \dots, X_{n+1}/T_{n+1}$ where $X_1, \dots, X_{n+1}$ are i.i.d. standard exponential random variables and $T_{n+1} := X_1 + \dots + X_{n+1}$. You may use this fact without proof.

Question 2: State and prove a CLT to accompany (1).

Question 3: Let $M_n := \max(Y_1, \dots, Y_{n+1})$ be the maximum of the spacings for a sample of size n . Show that $M_n \rightarrow_P 0$ as $n \rightarrow \infty$.

For the rest of this problem, we assume the following setup. Let $J = (x_0, \infty)$ be a subinterval of $\mathbb{R} = (-\infty, \infty)$, with x_0 finite or $-\infty$. Let f be a continuous, strictly positive probability density on J , $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample from a distribution with density f , and $(X_{(1)}, \dots, X_{(n)})$ be the order statistics for the sample $\mathbf{X}$.

Question 4: Suppose the above setup holds. Let k be a given positive integer less than n and let $y \in J$. Show that the conditional joint distribution of $(X_{(k+1)}, \dots, X_{(n)})$ given $X_{(k)} = y$ is the same as the joint distribution of

the order statistics of a random sample of size $n - k$ from the density

$$f_y(x) := \frac{f(x)I_{(y,\infty)}(x)}{\int_y^\infty f(t)dt},$$

where

$$I_{(y,\infty)}(x) := \begin{cases} 1, & y < x < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Note: Even if you were unable to answer Question 4, you should make use of it in the final part of this problem.

Question 5: For this question suppose that the setup of the problem holds with $J = (0, \infty)$ and $f(x) = \exp(-x)I_{(0,\infty)}(x)$, the density of the unit-rate exponential distribution. Let $1 \leq j \leq n$ be given. Find the distribution of

$$S_{n,j} := \sum_{k=j+1}^n (X_{(k)} - X_{(j)}).$$

Problem 2

Consider a simple symmetric random walk starting at 0 on $\mathbb{Z}$, in other words, $S_0 = 0$, $S_n = \sum_{i=1}^n X_i$ for $n \geq 1$ where $X_1, X_2, \dots$ are i.i.d. random variables uniformly distributed on $\{-1, 1\}$.

PART 1

Question 1: Show that $\{S_n\}$ is a Markov chain. Specify its state space and transition probability distribution.

Question 2: Does $\{S_n\}$ have a stationary distribution? Prove your answer.

Question 3: For any state x , show that

$$\mathbb{P}(S_n = x \text{ for infinitely many } n) = 1.$$

PART 2

For the remaining questions, let A be a fixed constant, and N be the stopping time

$$N := \min\{n : S_n \geq \frac{n}{2} + A\},$$

with $N := \infty$ if $S_n < \frac{n}{2} + A$ for every n . Our target is to prove

$$\mathbb{P}(N < \infty) \leq e^{-A}.$$

Question 4: Let $Z_n := e^{S_n - \frac{n}{2} - A}$. Prove that Z_n is a supermartingale.

Question 5: Prove that $\mathbb{P}(S_n \geq \frac{n}{2} + A \text{ for some } n \leq k) \leq e^{-A}$. (Hint: Use Markov's inequality. It may be helpful to define $N_k := \min\{N, k\}$.)

Question 6: Prove that $\mathbb{P}(N < \infty) \leq e^{-A}$.